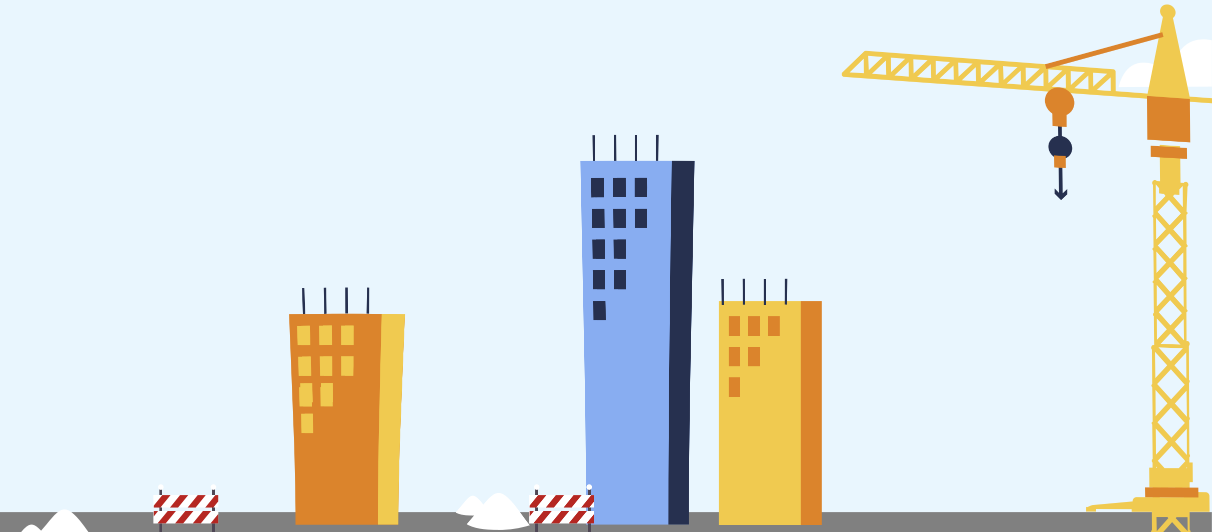
Sheild

Contributing to #MyBodyMyPPE

Increasing Safety and Participation of Women in Heavy Industry



*PPE: Personal Protective Equipment

Meet Our Team!

Jana
Abedelmaguid

Chengyi
Cai

Shaurya
Gulati

Seon
Jhang

Maryam
Khodadadi

Women in Heavy Industries

29%
Manufacturing

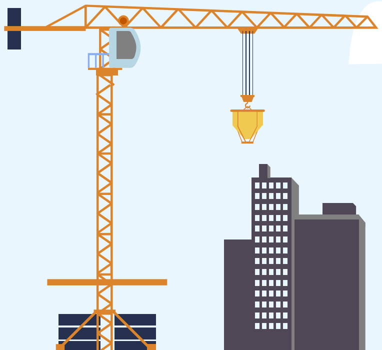
Source:
Manufacturing Institute

15%
Mining

Source:
The Bureau of Labor Statistics

11%
Construction

Source:
The Bureau of Labor Statistics



Current Status

70%

Lack access to properly fitted PPE

66%

Hindered on the job due to ill-fitted PPE

37%

Experienced a near-miss incident by ill-fitted PPE

Source: Survey on more than 500 women in construction, manufacturing, and other heavy industries
https://ehsdailyadvisor.blr.com/2025/03/new-report-highlights-inequities-in-womens-ppe/

Why Now?

Potential Market Growth in Women PPE

8.9%

2025 Year-over-Year

11.1%

CAGR 2024-2029

USD 14,293 M

Incremental Growth between 2024-2029

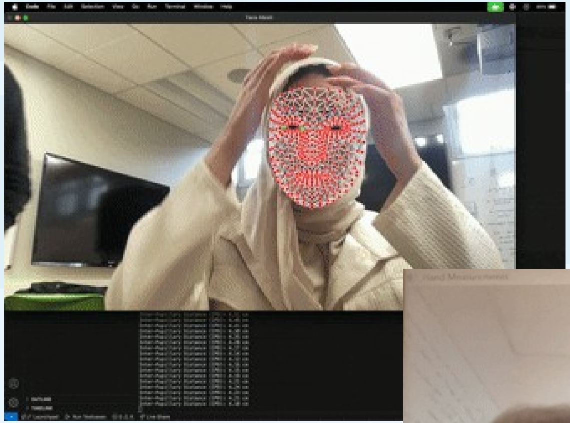
Source: Marketing Research by technavio https://www.technavio.com/report/personal-protective-equipment-ppe-market-for-women-analysis#:~:text=The%20personal%20protective%20equipment%20(PPE,the%20PPE%20market%20for%20women

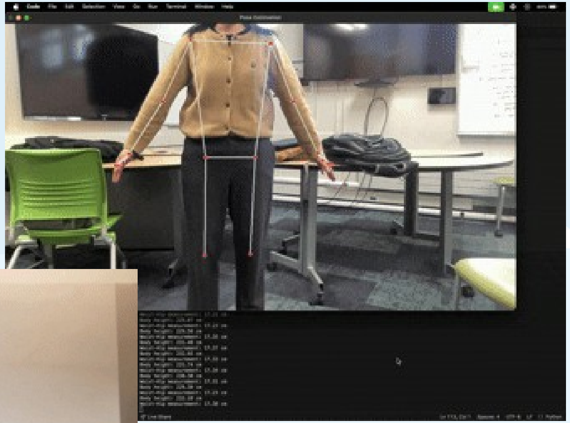
Our Objective

**Build a bridge between
industrial workplaces and manufacturers,
ensuring better-fitting and safer PPEs**

Our Solution

Using computer vision techniques to extract body measurements

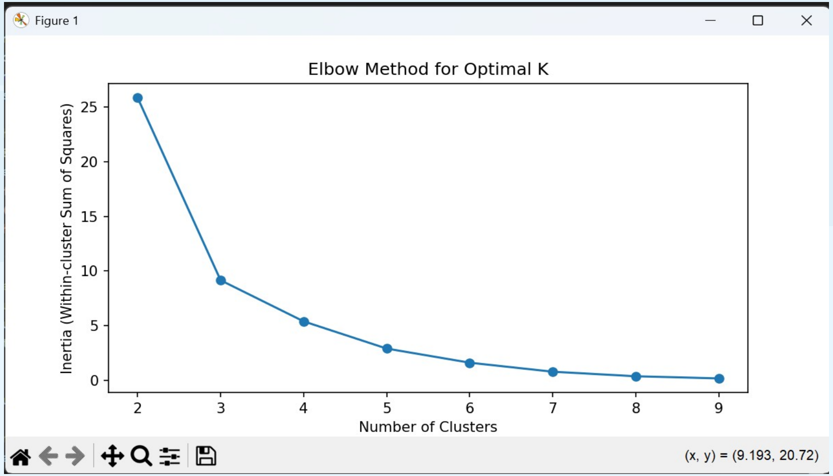




Our Solution

K-means clustering to identify and create groups

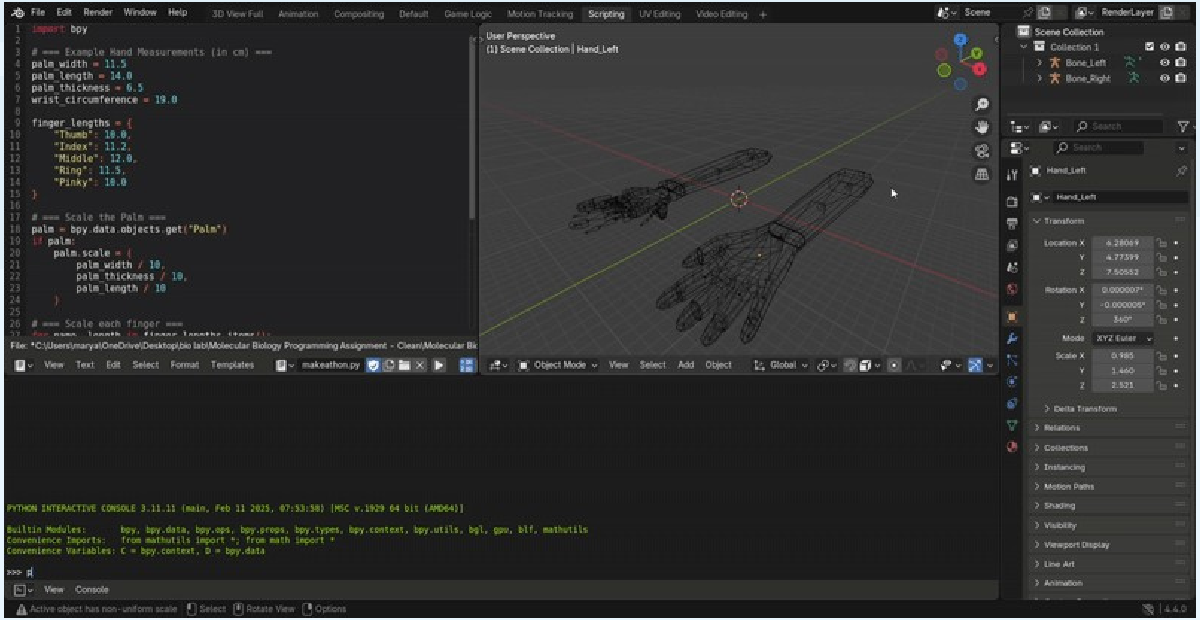
```python
for k in K_range:
    kmeans = KMeans(n_clusters=k, random_state=42, n_init=10)
    kmeans.fit(scaled_data)
    inertia.append(kmeans.inertia_)
    silhouette_scores.append(silhouette_score(scaled_data, kmeans.labels_))

# Plot the Elbow Method
plt.figure(figsize=(8,4))
plt.plot(K_range, inertia, marker="o")
plt.xlabel("Number of Clusters")
plt.ylabel("Inertia (Within-cluster Sum of Squares)")
plt.title("Elbow Method for Optimal K")
plt.show()

# Choose optimal K (e.g., from Elbow Method or Silhouette Score)
optimal_k = 3  # Adjust based on the elbow method
kmeans = KMeans(n_clusters=optimal_k, random_state=42, n_init=10)
clusters = kmeans.fit_predict(scaled_data)

# Add cluster labels to data
data["cluster"] = clusters

# Analyze cluster centers (average measurements per group)
cluster_centers = scaler.inverse_transform(kmeans.cluster_centers_)
cluster_df = pd.DataFrame(cluster_centers, columns=data.columns[:-1])
print(cluster_df)

# Save clustered data for 3D modeling
data.to_csv("clustered_hand_sizes.csv", index=False)
```

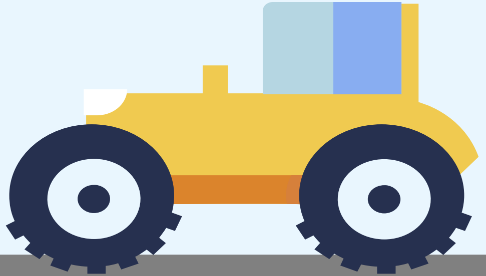


	Palm Width	Index Finger	Middle Finger	Ring Finger	Pinky Finger	Knuckle Width	Palm Circumference	Wrist Circumference
0	9.100000	8.35	9.05	8.600000	7.100000	9.900000	20.600000	16.250000
1	7.733333	7.00	7.70	7.233333	6.166667	8.466667	17.766667	14.433333
2	8.480000	7.76	8.46	8.020000	6.700000	9.220000	18.980000	15.280000

Our Solution

Using Blender to create 3D models



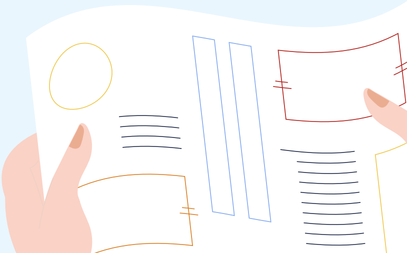


Improvements

We need:

- industry professionals’ opinion on what measurements are needed for various PPEs.

- better scale for our measurements, with a reference object, or a depth camera.



What's Next?

Inspiring young girls

XR Development

Students can have a 3D printed tool kits, like welding guns, and their XR would look something like this:


Thank You